Basic Cryptography and Hashing Tool using Python

1. Introduction
Cryptography is the practice of securing information by transforming it into an unreadable format. This project demonstrates basic encryption, decryption, hashing, and secure token generation using Python.

2. Objective
The objective of this project is to protect data, ensure confidentiality, maintain integrity, and demonstrate secure communication using basic cryptographic techniques.

3. Technologies Used
Python programming language with libraries such as hashlib, base64, and secrets.

4. Features
1. Text Encryption
2. Text Decryption
3. Hash Generation
4. Secure Token Generation
5. Menu-driven Interface

5. Methodology
The system uses Base64 encoding for encryption and decryption, SHA-256 for hashing, and secrets module for generating secure tokens.

6. Sample Code
Encryption: base64.b64encode()
Hashing: hashlib.sha256()
Token: secrets.token_hex()

7. Output
The program displays encrypted text, decrypted text, hash values, and secure tokens based on user input.

8. Conclusion
This project successfully demonstrates basic cryptographic operations using Python and highlights the importance of data security.